

Flat Tyre, Full Slot

Classifying Campus Bike-Rack Dwell Time Against Facilities' Daily Occupancy Photo

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Background

Facilities expands or removes a bike rack on the strength of one daily walkthrough photo, which counts every occupied slot the same way. Does a photo that can't tell a five-minute-old lock from a semester-old one measure use, or just presence?

Aims

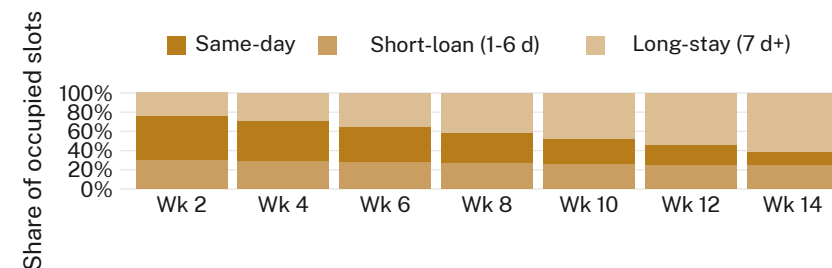
- Classify dwell time at every occupied slot across sampled racks
- Compare the classified dwell distribution to the walkthrough's single occupancy count
- Estimate what share of measured "utilisation" is long-stay abandonment

Methods

- 14 racks · 4-hourly timestamped photo log · 14-week deployment
- dwell classed from consecutive frames: same-day (<24 h), short-loan (1–6 d), long-stay (7 d+, unmoved)
- Facilities' own walkthrough schedule and counting rule held identical across both logs

Results

Long-stay bikes were under a quarter of tagged bikes at the outset but supplied most of the slots logged "occupied" by the deployment's final weeks; three of the four racks recommended for expansion this year sit above that trajectory's midpoint.



Long-stay bikes supplied a rising share of every week's occupied count, from 24% in week 2 to 61% by week 14, while the walkthrough's own daily photo recorded no change in total occupancy across the same span.

Discussion & conclusions

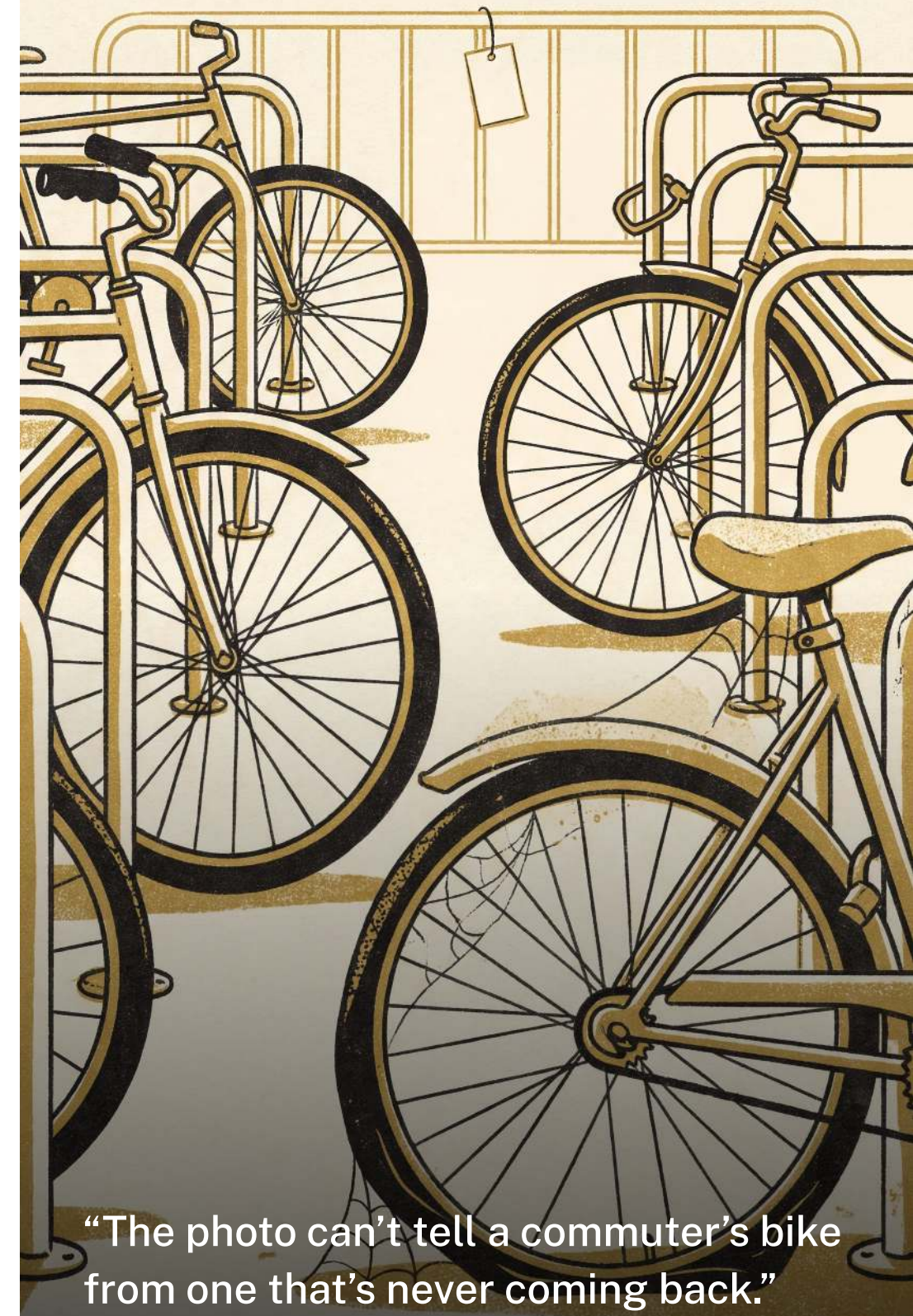
A slot photographed "occupied" isn't evidence anyone used the rack today. The walkthrough rewards racks that quietly accumulate long-stay bikes over ones that turn over fast. Next: pair the photo with a movement threshold before the next capital round.

References

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De-identified frame data only; no plate or lock serials retained. Long-stay flags rest on frame position alone, never bike ownership.

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“The photo can't tell a commuter's bike from one that's never coming back.”

61% of slots the daily walkthrough logged as "occupied" held a bike unmoved for a week or more by the deployment's close.